

Emmanuel Alabi

Software Engineer working toward AI & Robotics Research

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Research Interests

Embodied AI; robot learning; computer vision; autonomous systems; multimodal intelligence. Specifically, how an agent's representation of a physical environment should be shaped by having to move through it, and when learned components genuinely outperform well-specified classical methods.

Broader interests: machine learning, robotics, state estimation, planning, reinforcement learning.

Education

Babcock University — BSc (Hons) Software Engineering 2019 — 2023

Ilishan-Remo, Ogun State, Nigeria. School of Computing and Engineering Sciences.

Relevant coursework: Algorithms and Data Structures; Discrete Mathematics; Introductory Statistics; Introduction to Operations Research; Artificial Intelligence and Applications; Introduction to Big Data Engineering; Computer Organization and Assembly Language; Operating System I; Database System Design, Implementation and Management.

Research & Technical Experience

Independent Research Programme — Embodied & Multimodal Autonomy 2026 — present

- Self-directed programme building the mathematical, machine-learning and robotics foundations required to conduct independent research: linear algebra, probability, optimisation, state estimation and rigid-body geometry, each with implementations written from first principles.
- Reproducing published baselines in perception and navigation before extending them, on the principle that a reproduction which fails is itself a result worth documenting.
- Working toward a research question in embodied, multimodal autonomy precise enough to investigate with controlled experiments rather than demonstrations.
- Programme and progress documented publicly at emmanuelalabi.com/research/research-programme-2026

Selected Engineering Projects

Obelo — Multimodal AI Infrastructure

- Production AI platform combining multimodal generation, retrieval-augmented context, asynchronous orchestration, model routing and evaluation systems.
- Experimentation across multiple image-generation and language-model families; structured creative pipelines; reliability mechanisms for long-running AI workflows.
- emmanuelalabi.com/projects/obelo-multimodal-ai-infrastructure

Selected Technical Experience

Founder / Lead Engineer — Obelo

- Designed and developed a production AI platform: multimodal generation, retrieval-augmented brand context, model routing across model families, asynchronous orchestration, and evaluation and quality systems.

Software Engineer — SPay Business

- Production mobile-financial systems: authentication improvements, application performance, interface redesign, CI/CD automation and over-the-air mobile deployments.